PASSING AND TEAM SUCCESS IN SOCCER

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EDA

Data Selection

- From StatsBomb's Open Data project
- Modeled event data to analyze individual actions
- Focused on events classed as a 'pass' by StatsBomb
- Used lineup data for auxiliary information (ie. number of games each player was in)

Data Format & Cleaning

- Cleaned data manually from JSON to pandas DataFrame
- Encoded several columns with data for different event types (ie. passes, substitutions, etc.)
- Resulted in many sparse columns - first graph on the right visualizes this issue

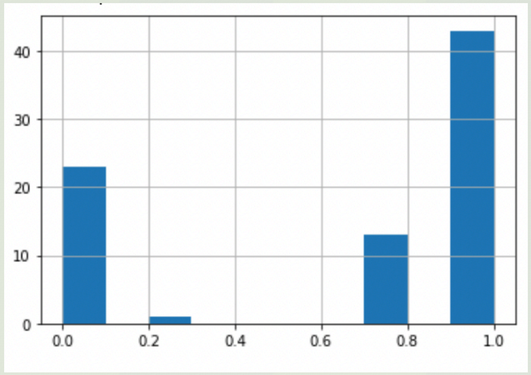
Data Visualization

- Created rough visualizations to use as model basis
- The first was of pass ending locations on the field (yellow points are where a team kept possession, red indicates end of possession)
- The second was of passing patterns for eleven selected positions in FC Barcelona across all provided matches

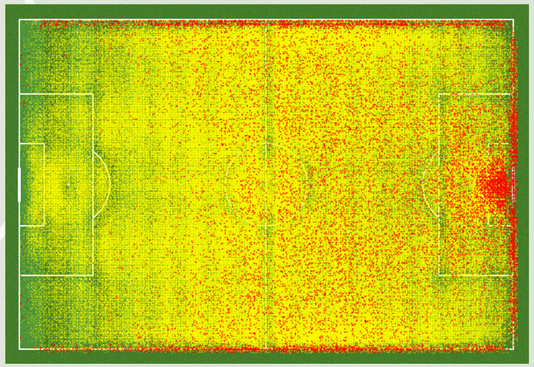
Acknowledgments

Pitch Visualization · MultiPage Streamlit · StatsBomb Open Data

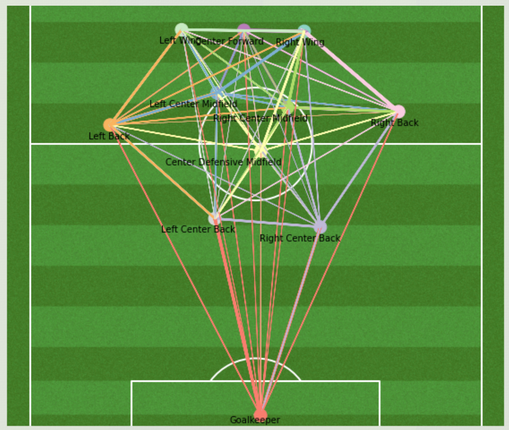
Number of Columns vs. Proportion of Null Values in Clean Data



Ends of Passes By Change in Possession



Position-Based Passmap for FC Barcelona



ABSTRACT

The use of data science in sports has exploded in recent years with teams and management in a variety of sports changing to a more data-driven decision making process in the hunt for glory. This project aims to continue that trend in soccer, and looks to analyze team success by examining their passing patterns and event potential over several years worth of data.



Steps	Tools Used
Cleaning	pandas & numpy
Exploratory Data Analysis	matplotlib, seaborn, & mplsoccer
Modeling	scikit-learn
Demo Creation	streamlit & Git LFS

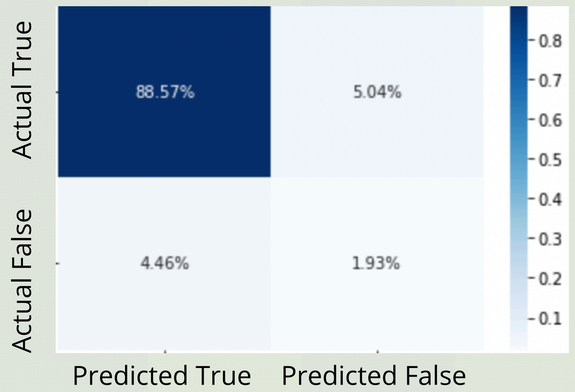
CONCLUSION

Our modeling process showed that there were identifiable patterns inside chains of passes. Teams can use their knowledge of these patterns to limit interruptions of passing chains and better their chances of moving towards the goal.

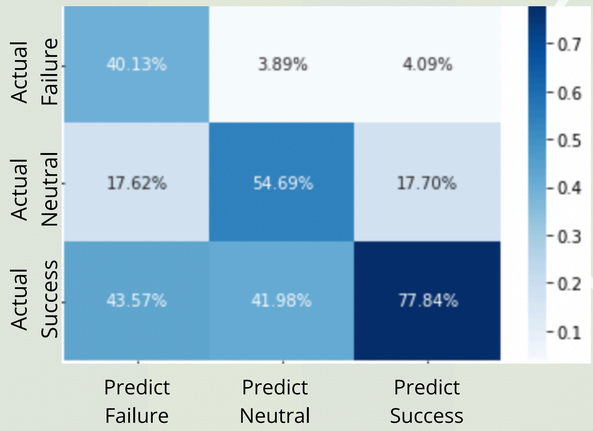
Limitations

- Dataset size - used Git LFS and Google Drive workarounds
- Team balance in dataset - majority of data was FC Barcelona (1 mil+ events)
- Visualizing complex location data

Confusion Matrix - Model 2



Confusion Matrix - Model 3



Future Analysis and Questions

- Individual player evaluation
- Inclusion of more varied global soccer data
- Connecting player & team evaluation with financial market evaluation
- How do these passing patterns turn into goal scoring opportunities?
- Ideal team composition (ie. what is the best possible version of Barcelona?)
- How do we model players or teams and their changes across seasons?

MODELING

Background

- Noted how often teams were making passes around the pitch
- Became interested in a way to "value" passes as a success or a fail.

<u>Three attempts:</u>
1) Does the pass stay in possession of the same team between events?
2) Does the pass eventually lead to a "Fail" or "Non-Fail" event?
3) Does the pass lead to a "Failure", "Success", or "Neutral" event? (Extension of Definition #2)

Process

- Performed data preprocessing, modeling, & optimization
- Used sklearn & skopt (sklearn-optimization) in Jupyter
- Scaled numerical data (height, angle, length)
- One-hot encoded categorical data (team name, pass style)
- Input our data into a Decision Tree Classifier - looking to make predictive results on a target variable

Results

	Model	Accuracy	F1-Score
0	Definition 1	0.97	0.95
1	Definition 2	0.91	0.95
2	Definition 3	0.70	0.67